20260401A Number Crunch

Background and Problem Intuition

Mathematical process: Start from Bayes' rule, then convert inference into optimisation.

$$p(\boldsymbol{\theta} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{y})} = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{\displaystyle\int p(\boldsymbol{y}, \boldsymbol{\theta})\, d\boldsymbol{\theta}} \tag{1}$$

$$\begin{aligned} p(\boldsymbol{y}) &= \int p(\boldsymbol{y}, \boldsymbol{\theta})\, d\boldsymbol{\theta} && (2)\\ &= \int p(\boldsymbol{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})\, d\boldsymbol{\theta}, && (3) \end{aligned}$$

where $\displaystyle\int p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$ is potentially intractable. To circumvent this, we introduce a tractable surrogate density $q(\boldsymbol{\theta})$ from a family of distributions $\mathcal{Q}$, and shift our focus to reformulating $\log p(\boldsymbol{y})$ as an expectation under $q(\boldsymbol{\theta})$. This converts the inference problem into an optimisation problem: instead of computing $p(\boldsymbol{\theta} \mid \boldsymbol{y})$ directly, we search for the $q(\boldsymbol{\theta}) \in \mathcal{Q}$ that best approximates it. This is made possible by noting that $\log p(\boldsymbol{y})$ contains no $\boldsymbol{\theta}$, so it is constant with respect to $\boldsymbol{\theta}$ and its value is unchanged by multiplication with any expression equal to 1. Since every probability density integrates to 1, both $\displaystyle\int p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$ and $\displaystyle\int q(\boldsymbol{\theta})\, d\boldsymbol{\theta}$ are valid choices, giving

$$\begin{aligned} \log p(\boldsymbol{y}) &= \log p(\boldsymbol{y}) \int p(\boldsymbol{\theta})\, d\boldsymbol{\theta} && (4)\\ &= \log p(\boldsymbol{y}) \int q(\boldsymbol{\theta})\, d\boldsymbol{\theta} && (5)\\ &= \int q(\boldsymbol{\theta}) \log p(\boldsymbol{y})\, d\boldsymbol{\theta} && (6)\\ &= \int q(\boldsymbol{\theta}) f(\boldsymbol{\theta})\, d\boldsymbol{\theta}, \quad f(\boldsymbol{\theta}) \equiv \log p(\boldsymbol{y}) && (7)\\ &= \mathbb{E}_q[f(\boldsymbol{\theta})] && (8)\\ & && (9) \end{aligned}$$

$$
\begin{aligned}
\log p(\boldsymbol{y}) &= \mathbb{E}_q[\log p(\boldsymbol{y})] && (10)\\
&= \mathbb{E}_q\left[\log \frac{p(\boldsymbol{y},\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \boldsymbol{y})}\right] && (11)\\
&= \mathbb{E}_q\left[\log \left(\frac{p(\boldsymbol{y},\boldsymbol{\theta})}{q(\boldsymbol{\theta})} \cdot \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \boldsymbol{y})}\right)\right] && (12)\\
&= \mathbb{E}_q\left[\log \frac{p(\boldsymbol{y},\boldsymbol{\theta})}{q(\boldsymbol{\theta})}\right] + \mathbb{E}_q\left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \boldsymbol{y})}\right] && (13)\\
&= \mathbb{E}_q[\log p(\boldsymbol{y},\boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] + \mathbb{E}_q\left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \boldsymbol{y})}\right]. && (14)
\end{aligned}
$$

Sequence of events (plain-language):

1. Start from $\log p(\boldsymbol{y})$, which is constant with respect to $\boldsymbol{\theta}$.
2. Multiply by 1 using normalisation of q: $\int q(\boldsymbol{\theta})\, d\boldsymbol{\theta} = 1$.
3. Rewrite as an integral: $\log p(\boldsymbol{y}) = \int q(\boldsymbol{\theta}) \log p(\boldsymbol{y})\, d\boldsymbol{\theta}$.
4. Recognise expectation form: $\int q(\boldsymbol{\theta}) f(\boldsymbol{\theta})\, d\boldsymbol{\theta} = \mathbb{E}_q[f(\boldsymbol{\theta})]$.
5. Use Bayes identity inside the logarithm: $p(\boldsymbol{y}) = \frac{p(\boldsymbol{y},\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\boldsymbol{y})}$.
6. Insert $q(\boldsymbol{\theta})/q(\boldsymbol{\theta})$ to create ELBO and KL terms.
7. Split product and quotient with log rules, then split expectations linearly.
8. Identify
$$\mathrm{ELBO}(q) = \mathbb{E}_q[\log p(\boldsymbol{y},\boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})]$$
and
$$\mathrm{KL}(q(\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta} \mid \boldsymbol{y})) = \mathbb{E}_q\left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \boldsymbol{y})}\right].$$

Define

$$
\begin{aligned}
\mathrm{ELBO}(q) &= \mathbb{E}_q[\log p(\boldsymbol{y},\boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] && (15)\\
&= \mathbb{E}_q[\log p(\boldsymbol{y},\boldsymbol{\theta})] + H(q), && (16)
\end{aligned}
$$

so that

$$\log p(\boldsymbol{y}) = \mathrm{ELBO}(q) + \mathrm{KL}(q(\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta} \mid \boldsymbol{y}))\,. \tag{17}$$

Since $\mathrm{KL}(\cdot\|\cdot) \geq 0$, maximising $\mathrm{ELBO}(q)$ is equivalent to minimising divergence to the true posterior.

Next Steps

Alignment with Course Requirements

Timeline

Summary